$\frac{-12}{5}$ on the number line? Between –2 and –3. Thus, a ratio made by dividing an integer by another integer is called a **rational number**. (Remember, we should not divide by zero!)

Formally speaking, a rational number is a number of the fractional (ratio) form $\frac{a}{b}$, where a and b are integers and $b \neq 0$. The collection of all rational numbers is denoted by $\mathbb{Q}$. Non-negative rational numbers may be thought of as fractions. They can also be expressed as decimals and percents.

Since operations on rational numbers demand a basic knowledge of operations on fractions, let us recall through an exercise some of the basic ideas related to fractions.

Recap Exercise

1. The simplest form of $\frac{125}{200}$ is _____.
2. Which of the following is not an equivalent fraction of $\frac{8}{12}$?
   (A) $\frac{2}{3}$ (B) $\frac{16}{24}$ (C) $\frac{32}{60}$ (D) $\frac{24}{36}$
3. Which is bigger: $\frac{4}{5}$ or $\frac{8}{9}$?
4. Add the fractions: $\frac{3}{5}+\frac{5}{8}+\frac{7}{10}$
5. Simplify: $\frac{1}{8}-\left(\frac{1}{6}-\frac{1}{4}\right)$
6. Multiply: $2\frac{3}{5}$ and $1\frac{4}{7}$.
7. Divide: $\frac{7}{36}$ by $\frac{35}{81}$.
8. Fill in the boxes: $\frac{\square}{66}=\frac{70}{\square}=\frac{28}{44}=\frac{\square}{121}=\frac{7}{\square}$
9. In a city $\frac{7}{20}$ of the population are women and $\frac{1}{4}$ are children. Find the fraction of the population of men.
10. Represent $\left(\frac{1}{2}+\frac{1}{4}\right)$ by a diagram.

Try these

1. Is the number –7 a rational number? Why?
2. Write any 6 rational numbers between 0 and 1.

Note

The word 'ratio' in Math refers to the comparison of the sizes of two different quantities of any kind. For example, if there is one teacher for every 20 students in a class, then the ratio of teachers to students is 1:20. Ratios are often written as fractions and so $1:20=\frac{1}{20}$. For this reason, numbers in the fractions form are called rational numbers.

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